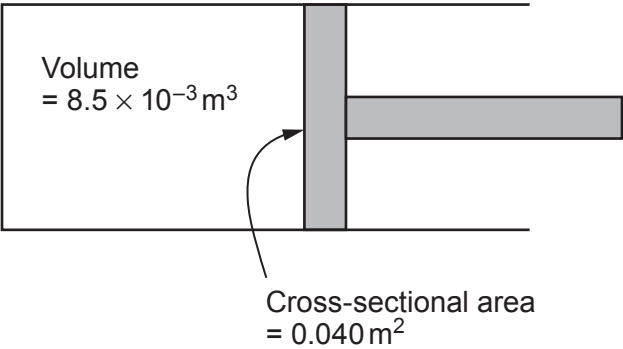


5. A sealed container with a leakproof piston of cross-sectional area 0.040 m^2 at one end has an initial volume of $8.5 \times 10^{-3}\text{ m}^3$. It contains oxygen gas at a pressure of $5.0 \times 10^5\text{ Pa}$ and a temperature of 285 K .

(Relative molecular mass of oxygen = 32)



(a) Calculate:

- (i) the number of moles of oxygen gas in the container; [2]

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- (ii) the number of molecules in the container; [1]

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- (iii) the rms speed of the molecules; [3]

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(iv) the force exerted by the gas on the piston. [1]

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(b) The gas is allowed to expand at a constant temperature of 285 K to a volume of $10.2 \times 10^{-3} \text{ m}^3$.

(i) Calculate the final pressure of the gas. [2]

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(ii) During the process the gas does 773 J of work. Determine the heat flowing into the gas, explaining your reasoning. [2]

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- (iii) Following the expansion the gas is returned to its initial state by a two-stage process:
- a decrease in volume at constant pressure;
 - an increase in pressure at constant volume.

A student says that approximately 60 J of heat flows into the gas during the overall cycle (the expansion and return of the gas to its original state). Justify this. Include a sketch of the paths on a p – V diagram with values on both axes. [4]

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